

Observe that:

Let $f: [a, b] \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, $\alpha, \beta \in [a, b]$. Suppose $f^{(n-1)}$ exists on $[a, b]$. Define

$$Q(t) = \frac{f(t) - f(\beta)}{t - \beta} \quad (t \neq \beta).$$

$$\text{Then } f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} \cdot (\beta - \alpha)^k + \frac{Q^{(n-1)}(\alpha)}{(n-1)!} \cdot (\beta - \alpha)^n$$

Proof. Derivating $(n-1)$ times to:

$$(t - \beta) \cdot Q(t) = f(t) - f(\beta)$$

$$\begin{aligned} \text{Then, } Q(t) + (t - \beta) Q'(t) &= f'(t) \\ Q'(t) + Q'(t) + (t - \beta) Q''(t) &= f''(t) \\ &\vdots \end{aligned}$$

$$(n-1) \cdot Q^{(n-2)}(t) + (t - \beta) \cdot Q^{(n-1)}(t) = f^{(n-1)}(t)$$

$$\text{Now, } f(\beta) = f(t) + Q(t) \cdot (\beta - t)$$

$$\begin{aligned} &= f(t) + f'(t) \cdot (\beta - t) + Q'(t) (\beta - t)^2 \\ &= f(t) + f'(t) (\beta - t) + \frac{f''(t)}{2!} (\beta - t)^2 + \frac{Q''(t)}{2!} (\beta - t)^3 \\ &\quad \vdots \\ &= \sum_{k=0}^{n-1} \frac{f^{(k)}(t)}{k!} \cdot (\beta - t)^k + \frac{Q^{(n-1)}(t)}{(n-1)!} \cdot (\beta - t)^n. \end{aligned}$$

$$g(t) = (\beta - t)^n$$

$$\text{Consider } F(t) = \sum_{k=0}^{n-1} \frac{Q^{(k)}(t)}{k!} (\beta - t)^k \quad \text{and} \quad G(t) = \sum_{k=0}^{n-1} \frac{g^{(k)}(t)}{k!} (\beta - t)^k$$

$$\begin{aligned} \text{Then, } F'(t) &= Q'(t) - Q^{(1)}(t) + Q^{(2)}(t) (\beta - t) \\ &\quad - \frac{1}{2!} \cdot 2 \cdot Q^{(2)}(t) \cdot (\beta - t) + \frac{1}{2!} \cdot Q^{(3)}(t) \cdot (\beta - t)^2 \\ &\quad - \frac{1}{3!} \cdot 3 \cdot Q^{(3)}(t) (\beta - t)^2 + \frac{1}{3!} Q^{(4)}(t) \cdot (\beta - t)^3 \\ &\quad \vdots \\ &\quad - \frac{1}{(n-1)!} \cdot (n-1) \cdot Q^{(n-1)}(t) (\beta - t)^{n-2} + \frac{1}{(n-1)!} \cdot Q^{(n)}(t) (\beta - t)^{n-1} \\ &= \frac{Q^{(n)}(t)}{(n-1)!} \cdot (\beta - t)^{n-1} \end{aligned}$$

Now, by the Generalized Mean-Value Theorem,

$$F'(s) \cdot [F(\alpha) - F(\beta)] = G'(s) [G(\alpha) - G(\beta)]$$

Let f be a map.

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(\alpha)}{k!} (x-\alpha)^k = f(\alpha) + \frac{f'(\alpha)}{1!} (x-\alpha) + \frac{f''(\alpha)}{2!} (x-\alpha)^2 + \dots + \frac{f^{(n)}(\alpha)}{n!} (x-\alpha)^n.$$

$$\left. \begin{array}{l} p_n(\alpha) = f(\alpha) \\ p_n'(\alpha) = f'(\alpha) \\ \vdots \\ p_n^{(n)}(\alpha) = f^{(n)}(\alpha) \\ p_n^{(n+1)}(\alpha) = 0 \\ \vdots \end{array} \right| \text{ Let } g_n(x) = (x-\alpha)^{n+1}. \text{ Then,}$$